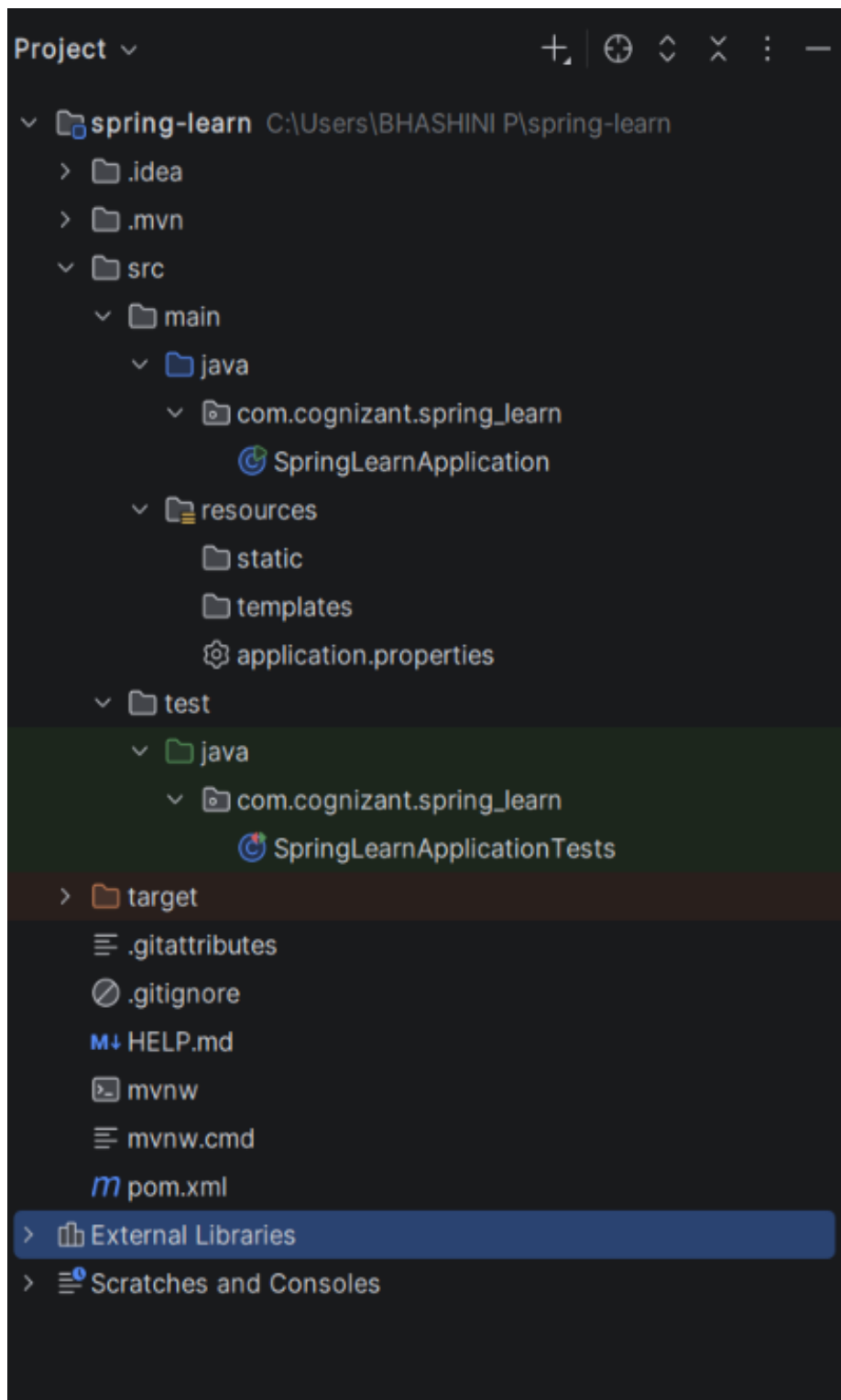


Spring REST using Spring Boot 3

1. spring-rest-handson - Create a Spring Web Project using Maven

Structure:



pom.xml:

```
SpringLearnApplication.java pom.xml (spring-learn) application.properties

1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
4     <modelVersion>4.0.0</modelVersion> Compatible with Maven 3
5     <parent>
6         <groupId>org.springframework.boot</groupId>
7         <artifactId>spring-boot-starter-parent</artifactId>
8         <version>4.1.0</version>
9         <relativePath/> <!-- lookup parent from repository -->
10    </parent>
11    <groupId>com.cognizant</groupId>
12    <artifactId>spring-learn</artifactId>
13    <version>0.0.1-SNAPSHOT</version>
14    <name/>
15    <description/>
16    <url/>
17    <licenses>
18        <license/>
19    </licenses>
20    <developers>
21        <developer/>
22    </developers>
23    <scm>
24        <connection/>
25        <developerConnection/>
26        <tag/>
27        <url/>
28    </scm>
29    <properties>
30        <java.version>21</java.version>
31    </properties>
32    <dependencies>
33        <dependency>
34            <groupId>org.springframework.boot</groupId>
35            <artifactId>spring-boot-starter-webmvc</artifactId>
36        </dependency>
37
38        <dependency>
39            <groupId>org.springframework.boot</groupId>
40            <artifactId>spring-boot-devtools</artifactId>
41            <scope>runtime</scope>
42            <optional>true</optional>
43        </dependency>
44        <dependency>
45            <groupId>org.springframework.boot</groupId>
46            <artifactId>spring-boot-starter-webmvc-test</artifactId>
47            <scope>test</scope>
48        </dependency>
49    </dependencies>
50    <build>
51        <plugins>
52            <plugin>
53                <groupId>org.springframework.boot</groupId>
54                <artifactId>spring-boot-maven-plugin</artifactId>
55            </plugin>
56        </plugins>
57    </build>
58 </project>
59
```

SpringLearnApplication.java :

```

1 package com.cognizant.spring_learn;
2
3 > import ...
4
5
6
7
8 @SpringBootApplication
9 public class SpringLearnApplication {
10
11     private static final Logger LOGGER = 2 usages
12         LoggerFactory.getLogger(SpringLearnApplication.class);
13
14 public static void main(String[] args) {
15
16     LOGGER.info("Spring Boot Application Started");
17
18     SpringApplication.run(SpringLearnApplication.class, args);
19
20     LOGGER.info("Application Running Successfully");
21 }
22

```

Console showing Spring Boot started successfully :

The screenshot shows a Windows Command Prompt window titled "SpringLearnApplication". The command executed is "C:\Program Files\Java\jdk-25\bin\java.exe ...". The output displays the standard Spring Boot startup sequence:

```

10:34:37.572 [main] INFO com.cognizant.spring_learn.SpringLearnApplication -- Spring Boot Application Started
10:34:37.884 [restartedMain] INFO com.cognizant.spring_learn.SpringLearnApplication -- Spring Boot Application Started

      ____              _
     / ___/____  _____  ____
    / __//___ \ / ___/ / ___/
   /_/ //___/ \/_/\___/_/\___/

:: Spring Boot ::                (v4.1.0)


2026-07-28T10:34:38.482+05:30 INFO 3512 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 25.0.1 with
2026-07-28T10:34:38.485+05:30 DEBUG 3512 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Running with Spring Boot v4.1.0, Spring v7.0.8
2026-07-28T10:34:38.486+05:30 INFO 3512 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile
2026-07-28T10:34:38.579+05:30 INFO 3512 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.'
2026-07-28T10:34:38.579+05:30 INFO 3512 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the
2026-07-28T10:34:39.725+05:30 INFO 3512 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8082 (http)
2026-07-28T10:34:39.744+05:30 INFO 3512 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-28T10:34:39.746+05:30 INFO 3512 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-28T10:34:39.792+05:30 INFO 3512 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1.233 seconds
2026-07-28T10:34:40.211+05:30 INFO 3512 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8082 (http) with context path '/'
2026-07-28T10:34:40.220+05:30 INFO 3512 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.33 seconds (process running...)
2026-07-28T10:34:40.235+05:30 INFO 3512 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Application Running Successfully

```

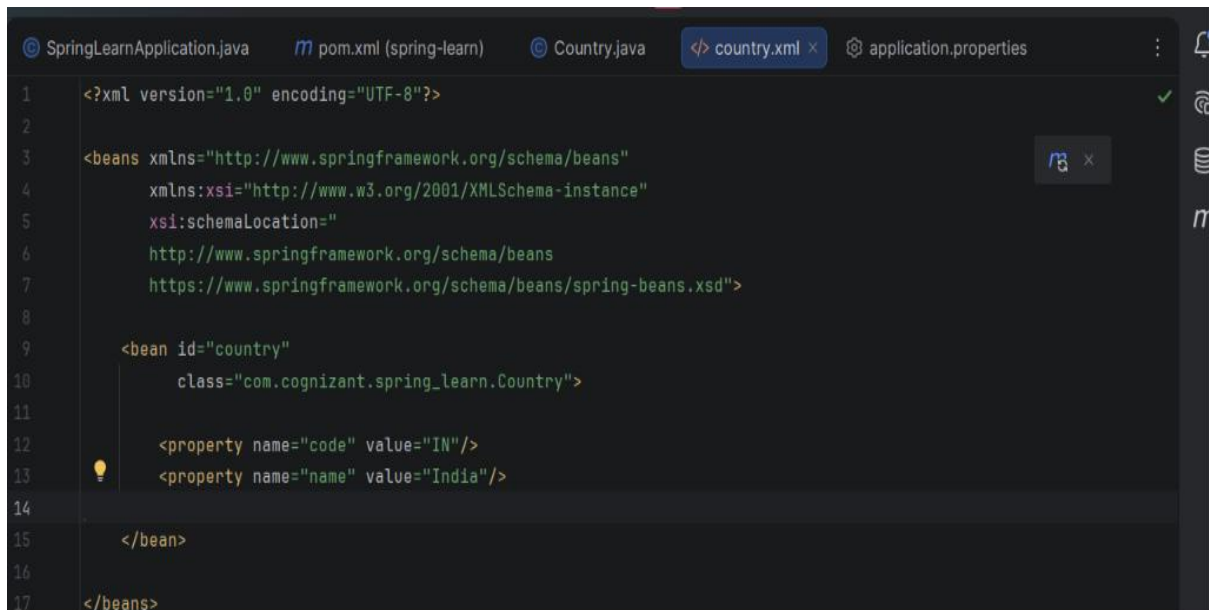
1. spring-rest-handson- Spring Core – Load Country from Spring Configuration XML (Files attached below)

Country.java

```
SpringLearnApplication.java pom.xml (spring-learn) Country.java x country.xml application.properties
1 package com.cognizant.spring_learn;
2 import org.slf4j.Logger;
3 import org.slf4j.LoggerFactory;
4
5 public class Country { 4 usages
6     private static final Logger LOGGER = 5 usages
7         LoggerFactory.getLogger(Country.class);
8
9     private String code; 3 usages
10    private String name; 3 usages
11    public Country() { no usages
12        LOGGER.debug("Inside Country Constructor.");
13    }
14    public String getCode() { no usages
15        LOGGER.debug("Inside getCode()");
16        return code;
17    }
18    public void setCode(String code) { no usages
19        LOGGER.debug("Inside setCode()");
20        this.code = code;
21    }
22    public String getName() { no usages
23        LOGGER.debug("Inside getName()");
24        return name;
25    }
26    public void setName(String name) { no usages
27        LOGGER.debug("Inside setName()");
28        this.name = name;
29    }
30 }
```

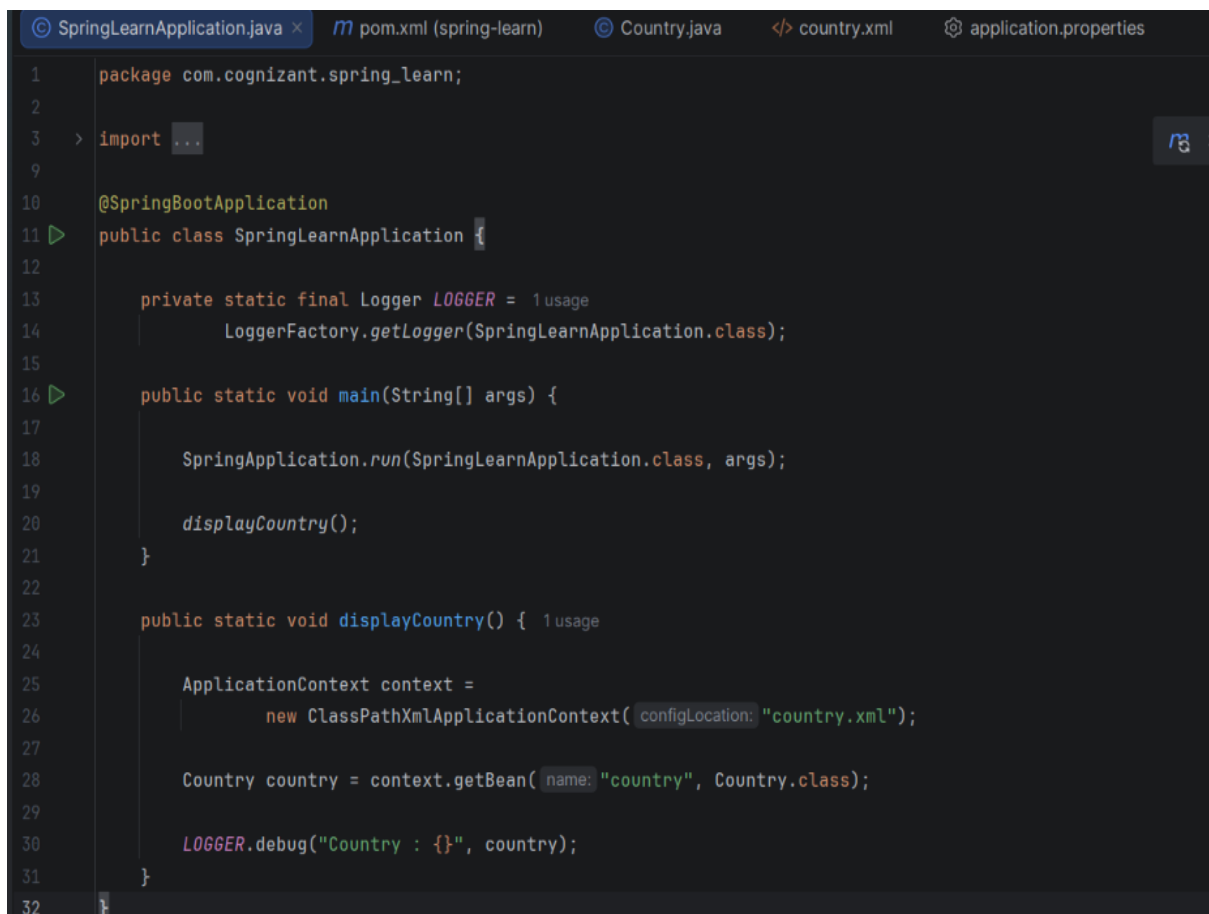
```
30
31 @Override
32 public String toString() {
33     return "Country{" +
34         "code='" + code + '\'' +
35         ", name='" + name + '\'' +
36         '}';
37 }
38 }
39
```

country.xml



```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9     <bean id="country"
10          class="com.cognizant.spring_learn.Country">
11
12         <property name="code" value="IN"/>
13         <property name="name" value="India"/>
14
15     </bean>
16
17 </beans>
```

SpringLearnApplication.java



```
1 package com.cognizant.spring_learn;
2
3 > import ...
4
5
6
7
8
9
10 @SpringBootApplication
11 public class SpringLearnApplication {
12
13     private static final Logger LOGGER = 1 usage
14         LoggerFactory.getLogger(SpringLearnApplication.class);
15
16     public static void main(String[] args) {
17
18         SpringApplication.run(SpringLearnApplication.class, args);
19
20         displayCountry();
21     }
22
23     public static void displayCountry() { 1 usage
24
25         ApplicationContext context =
26             new ClassPathXmlApplicationContext( configLocation: "country.xml");
27
28         Country country = context.getBean( name: "country", Country.class);
29
30         LOGGER.debug("Country : {}", country);
31     }
32 }
```

Console output

```
Run SpringLearnApplication x
"C:\Program Files\Java\jdk-25\bin\java.exe" ...

  ____  _
 / ___|| | | |
| |___| |_| |
 \___ \|  _/
      |_| |_|

:: Spring Boot ::                (v4.1.0)

2026-07-20T11:59:54.251+05:30 INFO 9312 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 25.0.1 with
2026-07-20T11:59:54.254+05:30 DEBUG 9312 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Running with Spring Boot v4.1.0, Spring v7.0.8
2026-07-20T11:59:54.255+05:30 INFO 9312 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile
2026-07-20T11:59:54.353+05:30 INFO 9312 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools'
2026-07-20T11:59:54.354+05:30 INFO 9312 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the
2026-07-20T11:59:54.737+05:30 INFO 9312 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8082 (http)
2026-07-20T11:59:54.757+05:30 INFO 9312 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-20T11:59:54.757+05:30 INFO 9312 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-20T11:59:54.813+05:30 INFO 9312 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1
2026-07-20T11:59:54.813+05:30 INFO 9312 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8082 (http) with context path '/'
2026-07-20T11:59:54.815+05:30 INFO 9312 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.724 seconds (process
2026-07-20T11:59:54.838+05:30 DEBUG 9312 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T11:59:54.840+05:30 DEBUG 9312 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T11:59:54.840+05:30 DEBUG 9312 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T11:59:54.842+05:30 DEBUG 9312 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Country : Country{code='IN', name='India'}
```

2. spring-rest-handson- Hello World RESTful Web Service (files attached below)

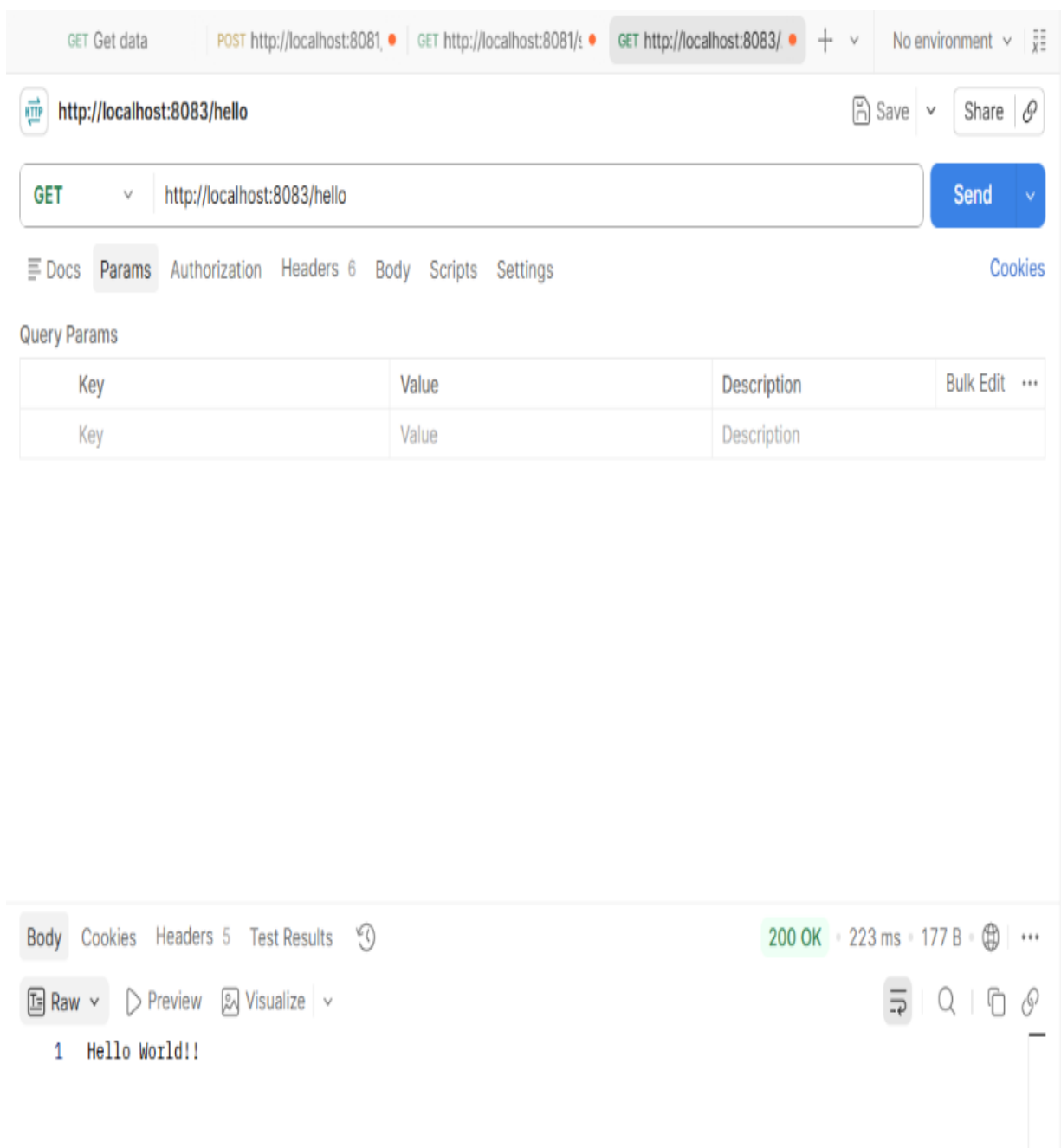
HelloController.java

```
tion.java pom.xml (spring-learn) Country.java country.xml application.properties HelloController.java x
1 package com.cognizant.spring_learn.controller;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.web.bind.annotation.GetMapping;
6 import org.springframework.web.bind.annotation.RestController;
7
8 @RestController
9 public class HelloController {
10
11     private static final Logger LOGGER = 2 usages
12         LoggerFactory.getLogger(HelloController.class);
13
14     @GetMapping("/hello")
15     public String sayHello() {
16
17         LOGGER.info("START");
18
19         LOGGER.info("END");
20
21         return "Hello World!!";
22     }
23 }
```

Browser output :



Postman response:



Console :

```
Run SpringLearnApplication x
(C) \_ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ |
| \_ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | | | | | | | | | | |
|   | _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ |
|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
=====|_|=====|_|_/_/_/

:: Spring Boot ::                (v4.1.0)

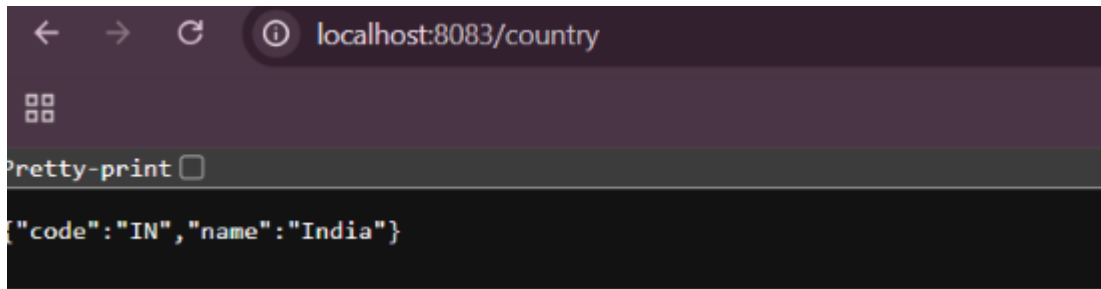
2026-07-20T12:16:23.699+05:30 INFO 15568 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 25.0.1 with
2026-07-20T12:16:23.703+05:30 DEBUG 15568 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Running with Spring Boot v4.1.0, Spring v7.0.8
2026-07-20T12:16:23.704+05:30 INFO 15568 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: 'default'
2026-07-20T12:16:23.775+05:30 INFO 15568 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.restart.enabled' to 'true' to enable restart
2026-07-20T12:16:23.775+05:30 INFO 15568 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'spring.devtools.web-console.enabled' property
2026-07-20T12:16:25.244+05:30 INFO 15568 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-20T12:16:25.384+05:30 INFO 15568 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-20T12:16:25.385+05:30 INFO 15568 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-20T12:16:25.495+05:30 INFO 15568 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed
2026-07-20T12:16:26.077+05:30 INFO 15568 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path ''
2026-07-20T12:16:26.083+05:30 INFO 15568 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.831 seconds (process running for 3.07)
2026-07-20T12:16:26.332+05:30 DEBUG 15568 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T12:16:26.333+05:30 DEBUG 15568 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T12:16:26.333+05:30 DEBUG 15568 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T12:16:26.334+05:30 DEBUG 15568 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Country : Country{code='IN', name='India'}
2026-07-20T12:16:48.687+05:30 INFO 15568 --- [spring-learn] [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-20T12:16:48.687+05:30 INFO 15568 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-20T12:16:48.687+05:30 INFO 15568 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms
2026-07-20T12:16:48.762+05:30 INFO 15568 --- [spring-learn] [nio-8083-exec-1] c.c.s.controller.HelloController : START
2026-07-20T12:16:48.762+05:30 INFO 15568 --- [spring-learn] [nio-8083-exec-1] c.c.s.controller.HelloController : END
```

2. spring-rest-handson – REST - Country Web Service

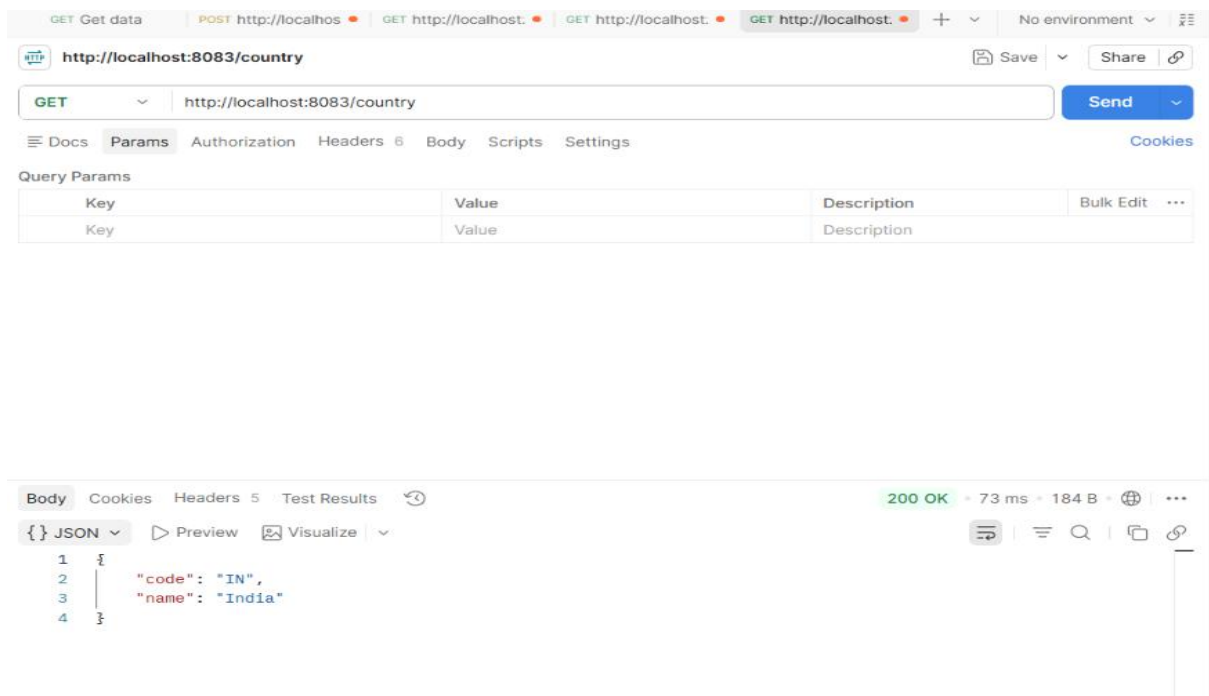
countryController.java

```
ng-learn) Country.java country.xml application.properties HelloController.java CountryController.java x
1 package com.cognizant.spring_learn.controller;
2 import com.cognizant.spring_learn.Country;
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5 import org.springframework.web.bind.annotation.RequestMapping;
6 import org.springframework.web.bind.annotation.RestController;
7
8 @RestController
9 public class CountryController {
10
11     @RequestMapping("/country")
12     public Country getCountryIndia() {
13
14         ApplicationContext context =
15             new ClassPathXmlApplicationContext("country.xml");
16
17         Country country = context.getBean("country", Country.class);
18
19         return country;
20     }
21 }
```

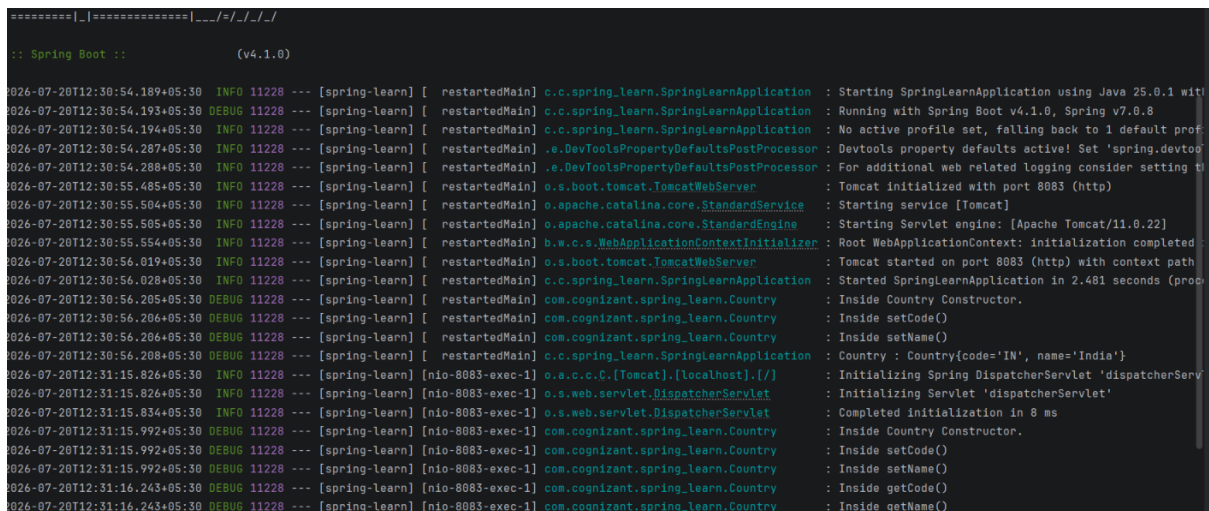

Browser output



Postman output



Console logs



2. Spring-rest-handson - REST - Get country based on country code(Attached screenshots of executions)

country.xml

```
SpringLearnApplication.java pom.xml (spring-learn) Country.java country.xml CountryService.java api
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9     <!-- Single Country Bean -->
10    <bean id="country" class="com.cognizant.spring_learn.Country">
11        <property name="code" value="IN"/>
12        <property name="name" value="India"/>
13    </bean>
14
15    <!-- Country List Bean -->
16    <bean id="countryList" class="java.util.ArrayList">
17        <constructor-arg>
18            <list>
19
20                <bean class="com.cognizant.spring_learn.Country">
21                    <property name="code" value="IN"/>
22                    <property name="name" value="India"/>
23                </bean>
24
25                <bean class="com.cognizant.spring_learn.Country">
26                    <property name="code" value="US"/>
27                    <property name="name" value="United States"/>
28                </bean>
29
30                <bean class="com.cognizant.spring_learn.Country">
31                    <property name="code" value="DE"/>
32                    <property name="name" value="Germany"/>
33                </bean>
34
35                <bean class="com.cognizant.spring_learn.Country">
36                    <property name="code" value="JP"/>
37                    <property name="name" value="Japan"/>
38                </bean>
39
40            </list>
41        </constructor-arg>
42    </bean>
43
44 </beans>
```

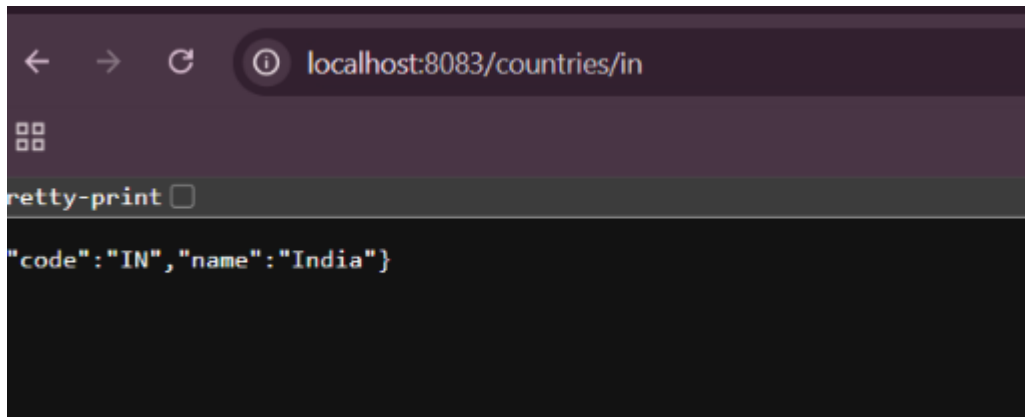
CountryController.java

```
va </> country.xml CountryService.java application.properties HelloController.java CountryController.java x v :
1 package com.cognizant.spring_learn.controller;
2
3 import com.cognizant.spring_learn.Country;
4 import com.cognizant.spring_learn.service.CountryService;
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.web.bind.annotation.*;
7
8 @RestController
9 public class CountryController {
10
11     @Autowired
12     private CountryService countryService;
13
14     @GetMapping("/countries/{code}")
15     public Country getCountry(@PathVariable String code) {
16
17         return countryService.getCountry(code);
18     }
19
20 }
```

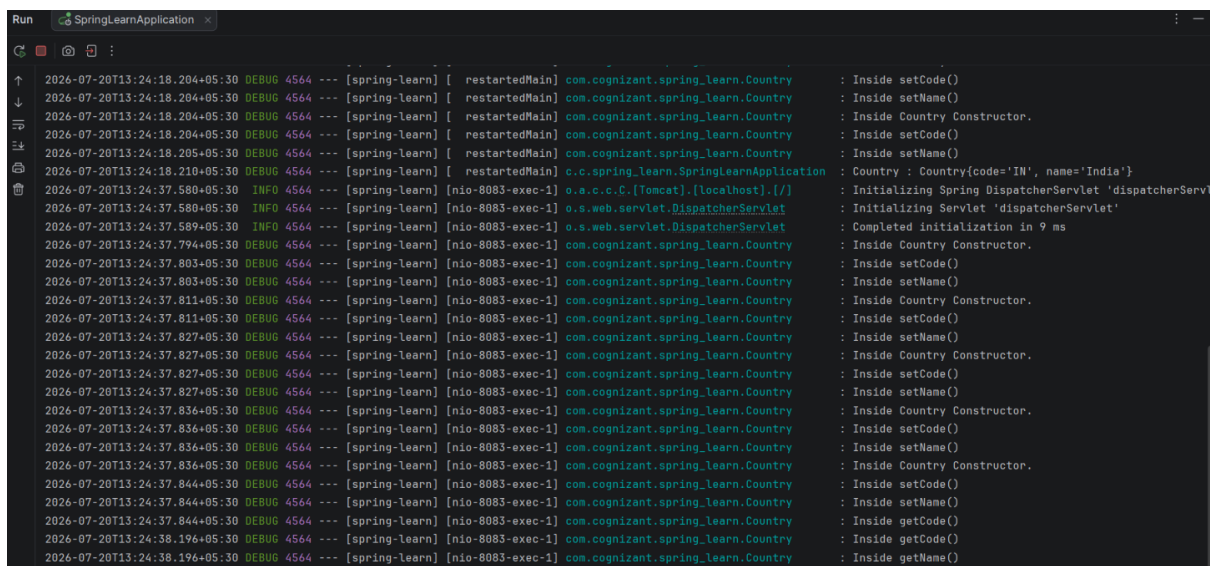
CountryService.java

```
va </> country.xml CountryService.java x application.properties HelloController.java CountryController.java v :
1 package com.cognizant.spring_learn.service;
2
3 import com.cognizant.spring_learn.Country;
4 import org.springframework.context.ApplicationContext;
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6 import org.springframework.stereotype.Service;
7
8 import java.util.List;
9
10 @Service
11 public class CountryService {
12
13     public Country getCountry(String code) { no usages
14
15         ApplicationContext context =
16             new ClassPathXmlApplicationContext("country.xml");
17
18         List<Country> countries =
19             (List<Country>) context.getBean("countrylist");
20
21         for (Country country : countries) {
22
23             if (country.getCode().equalsIgnoreCase(code)) {
24                 return country;
25             }
26         }
27
28         return null;
29     }
30 }
```

Browser response:



Console logs:

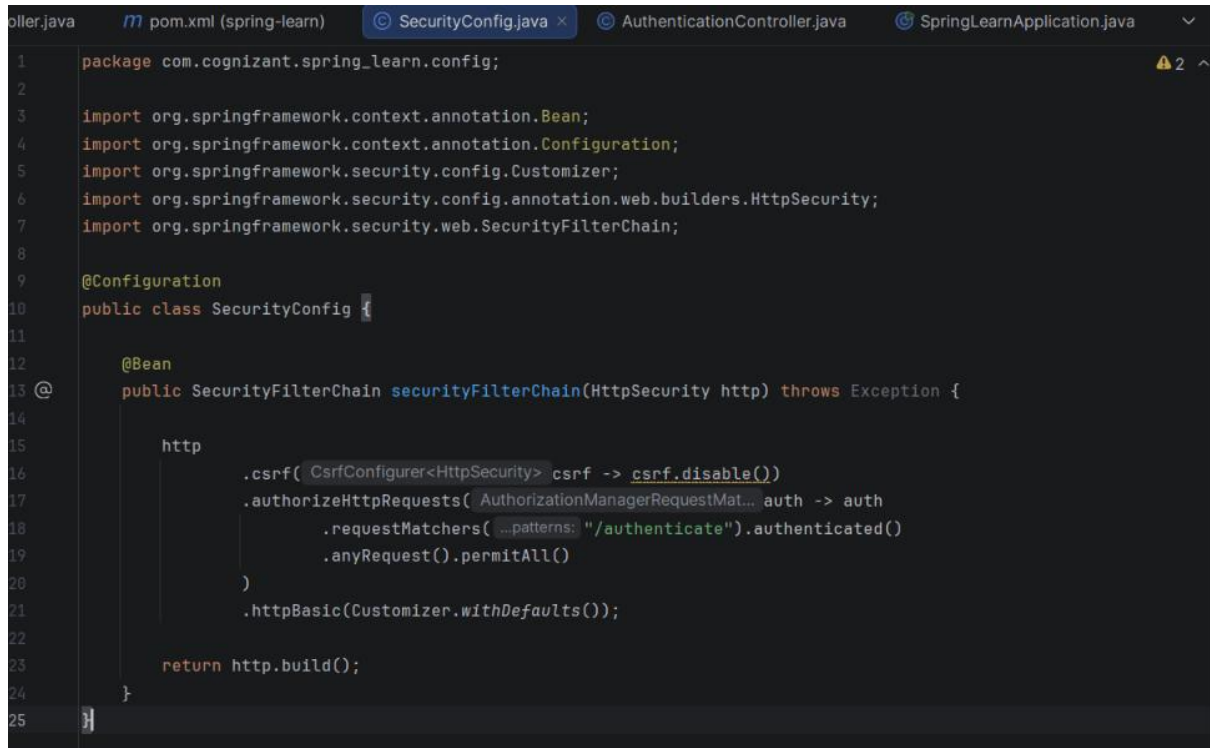


5. JWT-handson - Create authentication service that returns JWT

Step 1 : Create authentication controller and configure it in SecurityConfig.

Project Structure

- SecurityConfig.java



```
1 package com.cognizant.spring_learn.config;
2
3 import org.springframework.context.annotation.Bean;
4 import org.springframework.context.annotation.Configuration;
5 import org.springframework.security.config.Customizer;
6 import org.springframework.security.config.annotation.web.builders.HttpSecurity;
7 import org.springframework.security.web.SecurityFilterChain;
8
9 @Configuration
10 public class SecurityConfig {
11
12     @Bean
13     @Security
14     public SecurityFilterChain securityFilterChain(HttpSecurity http) throws Exception {
15
16         http
17             .csrf((CsrfConfigurer<HttpSecurity> csrf -> csrf.disable()))
18             .authorizeHttpRequests((AuthorizationManagerRequestMatcher<...> auth -> auth
19                 .requestMatchers(...patterns: "/authenticate").authenticated()
20                 .anyRequest().permitAll()
21             )
22             .httpBasic(Customizer.withDefaults());
23
24         return http.build();
25     }
26 }
```

- AuthenticationController.java



```
1 package com.cognizant.spring_learn.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 @RestController
7 public class AuthenticationController {
8
9     @GetMapping("/authenticate")
10     public String authenticate() {
11         return "Authentication Successful";
12     }
13 }
```

- IntelliJ Console

```

  \ \ / _ _ \ _ _ \ ( _ ) _ _ \ _ _ \ \ \ \ \ \
  ( _ ) _ _ \ _ _ \ _ _ \ _ _ \ \ \ \ \ \ \ \ \
  \ \ _ _ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \
  ^ \ _ _ \ _ _ \ \ \ \ \ \ \ \ \ \ \ \ \ \ \
  =====|=====|=====|=====|=====|
:: Spring Boot ::                (v4.1.0)

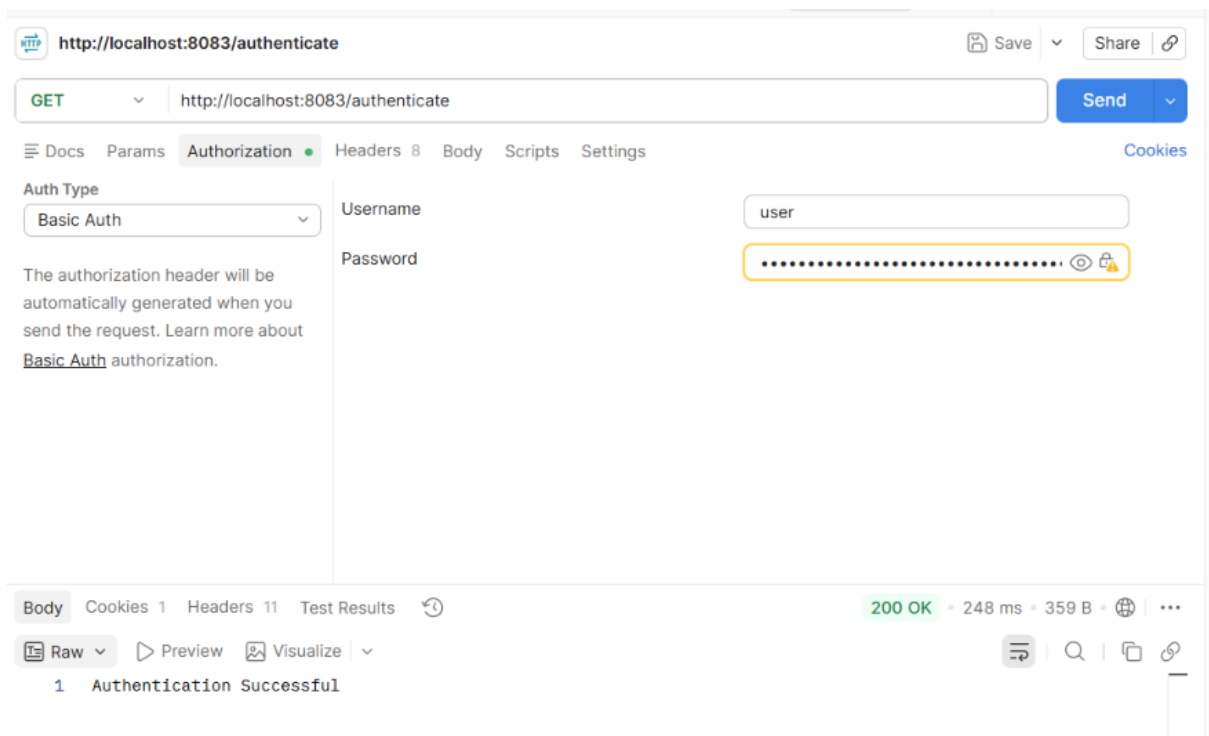
2026-07-20T13:50:56.668+05:30 INFO 15512 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 25.0.1 with
2026-07-20T13:50:56.677+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Running with Spring Boot v4.1.0, Spring v7.0.8
2026-07-20T13:50:56.678+05:30 INFO 15512 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default prof
2026-07-20T13:50:56.852+05:30 INFO 15512 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtool
2026-07-20T13:50:56.853+05:30 INFO 15512 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting tl
2026-07-20T13:50:58.477+05:30 INFO 15512 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-20T13:50:58.495+05:30 INFO 15512 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-20T13:50:58.495+05:30 INFO 15512 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-20T13:50:58.539+05:30 INFO 15512 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed
2026-07-20T13:50:58.768+05:30 WARN 15512 --- [spring-learn] [ restartedMain] .s.a.UserDetailsServiceAutoConfiguration :

Heaven generated security password: dKbHd4s-QeFQ-4644-01n0-19kaHh62e1nD
This generated password is for development use only. Your security configuration must be updated before running your application in production.

2026-07-20T13:50:58.777+05:30 INFO 15512 --- [spring-learn] [ restartedMain] r$InitializeUserDetailsServiceManagerConfigurer : Global AuthenticationManager configured with UserData
2026-07-20T13:50:59.260+05:30 INFO 15512 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path
2026-07-20T13:50:59.267+05:30 INFO 15512 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 3.432 seconds (proc
2026-07-20T13:50:59.426+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T13:50:59.428+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T13:50:59.428+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T13:50:59.429+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T13:50:59.429+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T13:50:59.430+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T13:50:59.430+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T13:50:59.430+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T13:50:59.430+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T13:50:59.431+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T13:50:59.431+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T13:50:59.431+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T13:50:59.431+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T13:50:59.431+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T13:50:59.431+05:30 DEBUG 15512 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()

```

- Postman



Step 2 : Read Authorization Header and Decode Username & Password

- AuthenticationController.java

```
ler.java pom.xml (spring-learn) SecurityConfig.java AuthenticationController.java x SpringLearnApplication.java
package com.cognizant.spring_learn.controller;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestHeader;
import org.springframework.web.bind.annotation.RestController;

import java.nio.charset.StandardCharsets;
import java.util.Base64;

@RestController
public class AuthenticationController {

    @GetMapping("/authenticate")
    public String authenticate(@RequestHeader("Authorization") String authHeader) {

        String encodedCredentials = authHeader.substring(beginIndex: 6);

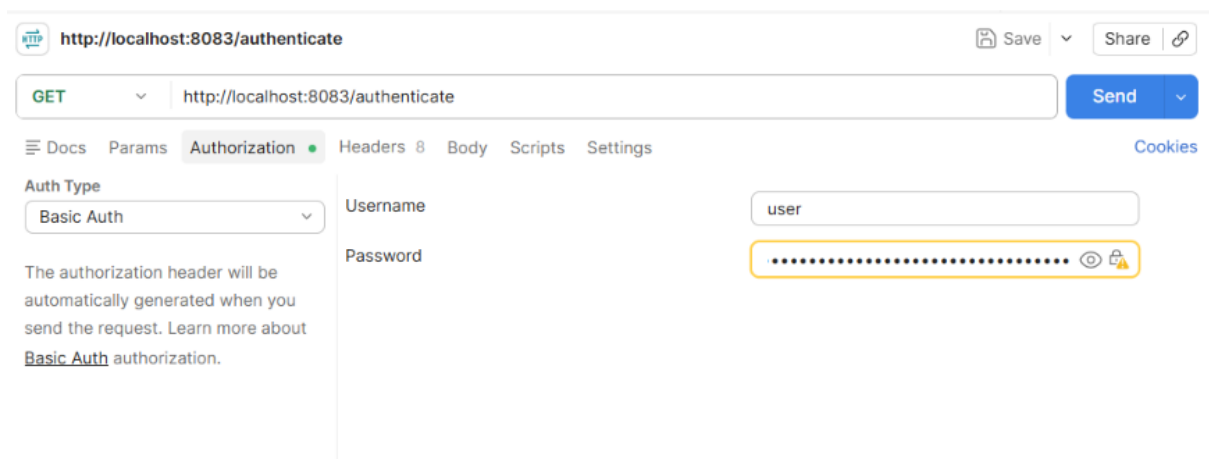
        byte[] decodedBytes = Base64.getDecoder().decode(encodedCredentials);
        String decodedCredentials = new String(decodedBytes, StandardCharsets.UTF_8);

        String[] values = decodedCredentials.split(regex: ":", limit: 2);

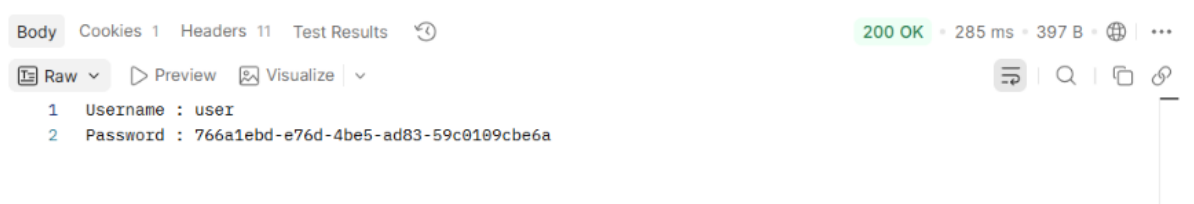
        String username = values[0];
        String password = values[1];

        return "Username : " + username + "\nPassword : " + password;
    }
}
```

- Postman Authorization



- Postman response :



- Spring Boot console :

```

SpringLearnApplication x
C:\Program Files\Java\jdk-25\bin\java.exe ...

  ____ _
 / ___ \| | | |
/ /   \| |_| |
\ \   /| | | |
 \___/\_|_|_|_|

:: Spring Boot ::      (v4.1.0)

2026-07-20T14:02:57.114+05:30 INFO 1900 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 25.0.1 with
2026-07-20T14:02:57.118+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Running with Spring Boot v4.1.0, Spring v7.0.8
2026-07-20T14:02:57.119+05:30 INFO 1900 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile
2026-07-20T14:02:57.222+05:30 INFO 1900 --- [spring-learn] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtool
2026-07-20T14:02:57.222+05:30 INFO 1900 --- [spring-learn] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the
2026-07-20T14:02:59.436+05:30 INFO 1900 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-20T14:02:59.459+05:30 INFO 1900 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-20T14:02:59.460+05:30 INFO 1900 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-20T14:02:59.518+05:30 INFO 1900 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 2.398 s
2026-07-20T14:02:59.928+05:30 WARN 1900 --- [spring-learn] [ restartedMain] s.a.UserDetailsServiceAutoConfiguration :

Using generated security password: 766a1ebd-e76d-4be5-ad83-59c0109cbe6a

This generated password is for development use only. Your security configuration must be updated before running your application in production.

2026-07-20T14:02:59.955+05:30 INFO 1900 --- [spring-learn] [ restartedMain] r.s.InitializeUserDetailsServiceConfigurer : Global AuthenticationManager configured with UserDetails
2026-07-20T14:03:00.727+05:30 INFO 1900 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-20T14:03:00.986+05:30 INFO 1900 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 4.699 seconds (process
2026-07-20T14:03:01.384+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T14:03:01.386+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T14:03:01.386+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T14:03:01.386+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T14:03:01.386+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T14:03:01.387+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T14:03:01.387+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T14:03:01.387+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T14:03:01.388+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T14:03:01.388+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T14:03:01.388+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T14:03:01.388+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T14:03:01.389+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-20T14:03:01.389+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-20T14:03:01.389+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-20T14:03:01.392+05:30 DEBUG 1900 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Country : Country{code='IN', name='India'}
2026-07-20T14:04:01.248+05:30 INFO 1900 --- [spring-learn] [nio-8083-exec-2] o.e.c.c.(Tomcat).[localhost]/ : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-20T14:04:01.248+05:30 INFO 1900 --- [spring-learn] [nio-8083-exec-2] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-20T14:04:01.257+05:30 INFO 1900 --- [spring-learn] [nio-8083-exec-2] o.s.web.servlet.DispatcherServlet : Completed initialization in 5 ms

```

Step 3 : Create authentication service that returns JWT.

AuthenticationController.java


```
SecurityConfig.java AuthenticationController.java StandardCharsets.java AuthenticationResponse.java JwtUtil
1 package com.cognizant.spring_learn.controller;
2
3 import com.cognizant.spring_learn.model.AuthenticationResponse;
4 import com.cognizant.spring_learn.util.JwtUtil;
5 import org.springframework.web.bind.annotation.GetMapping;
6 import org.springframework.web.bind.annotation.RequestHeader;
7 import org.springframework.web.bind.annotation.RestController;
8
9 import java.nio.charset.StandardCharsets;
10 import java.util.Base64;
11
12 @RestController
13 public class AuthenticationController {
14
15     @GetMapping("/authenticate")
16     public AuthenticationResponse authenticate(
17         @RequestHeader("Authorization") String authHeader) {
18
19         String encoded = authHeader.substring(beginIndex: 6);
20
21         byte[] decoded = Base64.getDecoder().decode(encoded);
22
23         String credentials = new String(decoded, StandardCharsets.UTF_8);
24
25         String username = credentials.split(regex: ":")[0];
26
27         String token = JwtUtil.generateToken(username);
28
29         return new AuthenticationResponse(token);
30     }
31 }
```

JwtUtil.java

```
AuthenticationController.java StandardCharsets.java AuthenticationResponse.java JwtUtil.java Sp
1 package com.cognizant.spring_learn.util;
2
3 import io.jsonwebtoken.Jwts;
4 import io.jsonwebtoken.security.Keys;
5
6 import javax.crypto.SecretKey;
7 import java.nio.charset.StandardCharsets;
8 import java.util.Date;
9
10 public class JwtUtil { 2 usages
11
12     private static final String SECRET = 1 usage
13         "mysecretkeymysecretkeymysecretkey12345";
14
15     private static final SecretKey KEY = 1 usage
16         Keys.hmacShaKeyFor(SECRET.getBytes(StandardCharsets.UTF_8));
17
18     public static String generateToken(String username) { 1 usage
19
20         return Jwts.builder()
21             .subject(username)
22             .issuedAt(new Date())
23             .expiration(new Date(System.currentTimeMillis() + 3600000))
24             .signWith(KEY)
25             .compact();
26     }
27 }
```

AuthenticationResponse.java

